

CSE 198 Week 7

Portfolio Optimization

RUDY OSUNA

May 14, 2026

Contents

1	Review: Why Build Portfolios?	3
1.1	Returns, Volatility, Annualization	3
2	Modern Portfolio Theory	3
2.1	The Efficient Frontier	3
2.2	The Capital Market Line and Sharpe	3
3	Why MPT Breaks in Practice	5
3.1	Estimation Error in μ	5
3.2	Estimating μ : Black-Litterman	5
3.3	Estimation Error in Σ	5
3.4	Time-Varying Distributions	6
3.5	The Symmetric Penalty of Variance	6
4	The Empirical Reality of Returns	6
4.1	CAPM Vocabulary: α and β	6
4.2	Skewness, Kurtosis, and Fat Tails	6
4.3	Stylized Facts of Returns	7
5	The Linear Algebra of Portfolios	7
5.1	Spectral Decomposition of Σ	7
5.2	Eigenportfolios	8
5.3	Principal Component Analysis (PCA) and Statistical Factor Models	8
5.4	The Spiked Covariance Model	9
5.5	Ledoit-Wolf Shrinkage	9
5.6	Risk Decomposition via Eigenportfolios	10
6	Better Risk-Adjusted Metrics	10
6.1	Sortino	10
6.2	Probabilistic Sharpe Ratio	10
6.3	Mean-CVaR	11
6.4	Which Metric to Use	11

7	Extra: Why Options Change the Math	11
8	Extra: Options Portfolio Optimization	12
9	What We Haven't Talked About	13

§1 Review: Why Build Portfolios?

A portfolio is a collection of positions in different assets held at the same time. We hold more than one asset because we do not know in advance which one will be the best, and even if we did, concentration exposes us to risks we could smooth out. The mathematical object behind “smoothing out” is **diversification**: if two assets are not perfectly correlated, blending them produces a portfolio whose volatility is strictly less than the weighted average of the component volatilities. This is the seed from which the entire lecture grows.

§1.1 Returns, Volatility, Annualization

Definition 1.1. The **simple return** over $[t-1, t]$ is $r_t = (P_t - P_{t-1})/P_{t-1}$, and the **log return** is $\tilde{r}_t = \log(P_t/P_{t-1})$.

Log returns are additive across time, so they are right for compounding. Simple returns are additive across assets in a single period, so they are right for portfolio math at a fixed time slice. The annualized volatility under i.i.d. daily returns is $\sigma_{\text{ann}} = \sigma_{\text{daily}}\sqrt{252}$, the square-root coming from variance scaling linearly with time. The annualized log return is exactly $252\mu_{\text{daily}}$; the annualized simple return is approximately $252\mu_{\text{daily}}$ for small daily returns.

A note on notation: σ is just the symbol for standard deviation. The harder question, which we return to throughout, is whether standard deviation is the right summary of risk in the first place. The answer is often yes, sometimes no, and almost always not the whole story.

§2 Modern Portfolio Theory

Modern Portfolio Theory (MPT), introduced by Markowitz (1952), formalizes portfolio construction as a constrained optimization. Given n assets with mean return vector $\mu \in \mathbb{R}^n$ and covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$, a portfolio is a weight vector w with $\sum w_i = 1$. Its expected return is $\mu_p = w^T \mu$ and its variance is $\sigma_p^2 = w^T \Sigma w$.

§2.1 The Efficient Frontier

Definition 2.1. A portfolio is **mean-variance efficient** if no other portfolio achieves strictly higher expected return at the same variance. The curve of efficient portfolios in (σ, μ) space is the **efficient frontier**.

For unconstrained MPT (long-short allowed, no other constraints), the frontier admits a closed form: the entire frontier is a hyperbola in (σ, μ) space parametrized by the target return.

§2.2 The Capital Market Line and Sharpe

Introducing a risk-free asset with return r_f , the efficient frontier becomes a straight line in (σ, μ) space, called the **Capital Market Line** (CML). It is

tangent to the risky-asset frontier at a single portfolio: the **tangency portfolio**, with weights proportional to $\Sigma^{-1}(\mu - r_f \mathbf{1})$.

Definition 2.2. The **Sharpe ratio** of a portfolio is $SR = (\mu_p - r_f)/\sigma_p$, the excess return per unit volatility.

Geometrically the Sharpe ratio is the slope of the line from $(0, r_f)$ to (σ_p, μ_p) , and the CML is the line of maximum slope. The tangency portfolio is therefore the unique portfolio of risky assets with the highest Sharpe.

Proposition 2.3 (Two-fund separation, Tobin 1958)

Every mean-variance efficient portfolio is a mix of the risk-free asset and the tangency portfolio. Investor preferences over (σ, μ) affect only the mixing weight, not the composition of the risky-asset bundle.

This is a clean result and a clean lie. It is clean because it says investors should solve one problem (the tangency portfolio) and slide along a line. It is a lie because we do not know μ or Σ .

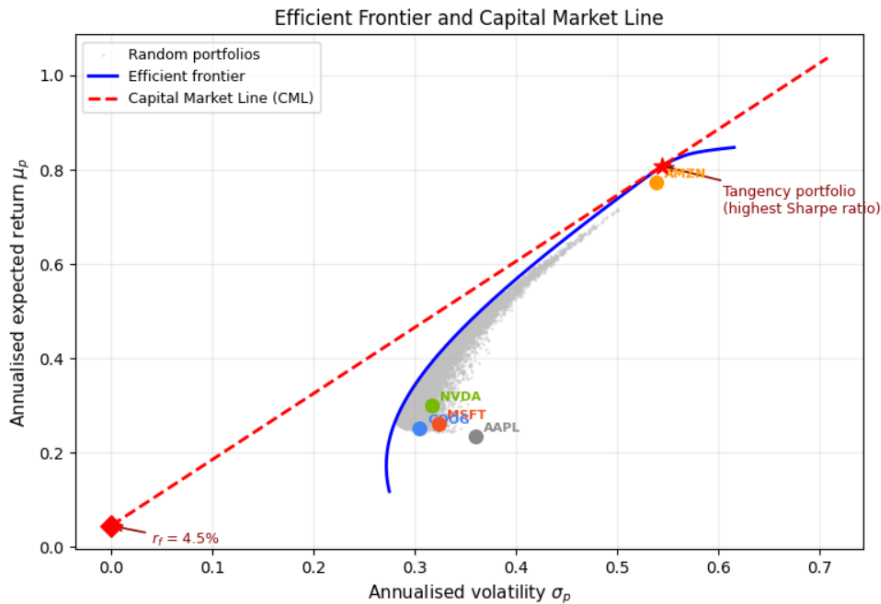


Figure 1: Efficient frontier and Capital Market Line for five large-cap stocks (NVDA, AAPL, MSFT, GOOG, AMZN), 2020–2024. The grey cloud is 15,000 random long-only portfolios. The blue curve is the analytical efficient frontier (long-short). The red dashed line is the CML tangent at the risk-free rate $r_f = 4.5\%$; the tangency portfolio achieves $\mu = 80.8\%$, $\sigma = 54.5\%$, Sharpe = 1.40. (See [implementation notebook](#))

§3 Why MPT Breaks in Practice

§3.1 Estimation Error in μ

To compute the tangency portfolio we need μ , the vector of expected returns. We never observe μ . We observe a finite sample and form an estimate $\hat{\mu}$. The variance of the sample mean of T observations is σ^2/T , so for an asset with annualized volatility 20% and $T = 252$ days of data, the standard error on the annual mean is $20\%/\sqrt{1} = 20\%$. To pin down a mean return to within 1% at this volatility we would need 400 years of data, which is more than we have for any asset.

The optimizer takes $\hat{\mu}$ and produces weights as $w \propto \Sigma^{-1}\hat{\mu}$. Small perturbations in $\hat{\mu}$ along the eigenvectors of Σ with small eigenvalues get amplified by Σ^{-1} into large changes in w . This is Michaud's famous critique:

Proposition 3.1 (Michaud 1989)

Unconstrained mean-variance optimization is an error-maximization algorithm: it places the largest bets precisely where the estimation error in μ has been amplified the most.

§3.2 Estimating μ : Black-Litterman

Black and Litterman (1992) propose a structured prior rather than a raw sample mean. The starting point is the return vector implied by market equilibrium: if w_{mkt} is mean-variance optimal, reverse-engineering gives $\Pi = \lambda \Sigma w_{\text{mkt}}$, where $\Pi \in \mathbb{R}^n$ is the vector of implied equilibrium excess returns and $\lambda > 0$ is a risk-aversion coefficient (not an eigenvalue). An investor expresses k views as $P\mu = Q$, where $P \in \mathbb{R}^{k \times n}$ picks the assets involved in each view, $Q \in \mathbb{R}^k$ gives the expected return for each view (not the orthogonal matrix of Section 5), and $\Omega \in \mathbb{R}^{k \times k}$ is a diagonal matrix of view uncertainties. The scalar $\tau > 0$ controls how much weight the prior receives relative to the views. The posterior is

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\Pi + P^T\Omega^{-1}Q].$$

The key property: with no views, you recover the market portfolio. With a strong view, you move toward it in proportion to your confidence. This stabilizes the optimizer because Π is well-conditioned by construction, it is backed by a portfolio millions of investors are already holding.

§3.3 Estimation Error in Σ

The sample covariance $S = \frac{1}{T-1} \sum (r_t - \bar{r})(r_t - \bar{r})^T$ is unbiased but for n assets has roughly $n^2/2$ free parameters estimated from nT data points. When n approaches T , S becomes nearly singular, its small eigenvalues are noise, and S^{-1} explodes. In the high-dimensional regime $n/T \rightarrow q \in (0, 1)$ the spectrum of S is not even close to that of Σ ; the discrepancy is described by the Marchenko-Pastur

law. Factor models and PCA address this by constraining Σ to a low-rank plus diagonal structure, reducing the parameter count dramatically (both topics covered in Section 5).

§3.4 Time-Varying Distributions

MPT assumes μ and Σ are constant. Real return distributions are not stationary. Nvidia (\$NVDA) in 2015 was a gaming-GPU company with one return distribution; \$NVDA in 2024 is an AI-infrastructure company with a different one. A single $\hat{\mu}$ estimated over a window straddling that transition is meaningless. Covariance has its own version of this problem: in calm markets correlations are moderate; in a crisis they all approach 1, so the diversification we paid for evaporates exactly when we needed it.

Note: The crisis “correlations going to one” phenomenon is a property of Σ , not of μ . People sometimes conflate the two because both shift in a crisis, but the math is different. Drift changes change which portfolios are optimal; correlation changes change how risky any given portfolio is.

§3.5 The Symmetric Penalty of Variance

Variance, and therefore Sharpe, penalizes upside and downside symmetrically. A portfolio with a large positive jump and one with a large negative jump of the same size contribute equally to σ . For most investors this is the wrong objective: nobody complains about volatility on the way up. The downside-only measures we discuss in Section 7 fix this.

§4 The Empirical Reality of Returns

§4.1 CAPM Vocabulary: α and β

The Capital Asset Pricing Model (Sharpe 1964) decomposes an asset’s return as

$$r_i - r_f = \alpha_i + \beta_i(r_m - r_f) + \varepsilon_i,$$

where r_m is the market return, $\beta_i = \text{Cov}(r_i, r_m) / \text{Var}(r_m)$ is the asset’s market sensitivity (a β of 2 means the asset moves twice as much as the market on average; $\beta = 0$ is market-neutral), α_i is the part of expected excess return that the market does not explain, and ε_i is the idiosyncratic residual with $\mathbb{E}[\varepsilon_i] = 0$ and $\text{Cov}(\varepsilon_i, r_m) = 0$. Alpha is what active managers claim to produce; beta is what an index fund gives you for cheap.

§4.2 Skewness, Kurtosis, and Fat Tails

Real returns are not Gaussian. Empirically, equity-return distributions are left-skewed (large negative moves are more common than equally large positive ones) and have excess kurtosis (the tails are fatter than a normal distribution would predict).

Definition 4.1. Let X be the return of a single asset with mean $\mu_X = \mathbb{E}[X]$ and standard deviation σ_X . The **skewness** is $\mathbb{E}[(X - \mu_X)^3]/\sigma_X^3$ and the **excess kurtosis** is $\mathbb{E}[(X - \mu_X)^4]/\sigma_X^4 - 3$. Both are zero for a Gaussian.

Negative skewness means the left tail is heavier than the right: large losses are more likely than equally large gains. Positive excess kurtosis means tails are fatter than a Gaussian predicts: extreme events happen more often than the bell curve says they should.

Fitting a normal distribution to \$NVDA daily returns makes this concrete (**credits: Quant Guild**). A four-sigma move has probability roughly 3.2×10^{-5} under the Gaussian, or one occurrence every 31,574 observations, about once every 125 years at daily frequency. In practice we observe multiple four-sigma days in NVDA's history without waiting a century; the model assigns them near-zero probability. The Gaussian is not just a little wrong here, it is wrong by orders of magnitude in the tails. Sharpe, which only uses the first two moments, sees none of this.

§4.3 Stylized Facts of Returns

A handful of empirical regularities recur across nearly every liquid asset. Raw returns show essentially no linear autocorrelation at daily horizons. But squared returns are strongly autocorrelated: big moves cluster in time, and past negative returns predict higher future volatility than positive ones of the same size. If returns were Gaussian and stationary, Markowitz would not break in practice.

§5 The Linear Algebra of Portfolios

The covariance matrix Σ has entries $\Sigma_{ij} = \text{Cov}(X_i, X_j)$, where X_i denotes the return of asset i . Since

$$\text{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] = \text{Cov}(X_j, X_i),$$

the matrix is symmetric. Its diagonal entries are $\Sigma_{ii} = \text{Var}(X_i)$, the variance of each asset. Because any variance is non-negative, Σ is also positive semidefinite.

§5.1 Spectral Decomposition of Σ

Theorem 5.1 (Spectral Theorem)

Every real-valued symmetric matrix $A \in \mathbb{R}^{n \times n}$ admits a decomposition $A = Q\Lambda Q^T$, where $Q \in \mathbb{R}^{n \times n}$ is orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_1 \geq \dots \geq \lambda_n$. The columns of Q are the eigenvectors of A and the diagonal entries of Λ are the corresponding eigenvalues.

Applied to the covariance matrix:

$$\Sigma = Q\Lambda Q^T,$$

with $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ (all non-negative because Σ is positive semidefinite). An orthogonal matrix has orthonormal columns: the eigenvectors q_i satisfy $q_i^T q_j = 0$ for $i \neq j$ and $\|q_i\| = 1$, so they are mutually perpendicular and unit-normalized.

§5.2 Eigenportfolios

Recall that a portfolio is a weight vector $w \in \mathbb{R}^n$. An **eigenportfolio** is simply a portfolio whose weight vector is an eigenvector of Σ , meaning $\Sigma q_i = \lambda_i q_i$ for some scalar λ_i . The composition of the portfolio happens to align with a natural risk axis of the covariance matrix.

Each eigenvector q_i is itself a (signed, possibly unnormalized) portfolio. Its variance under Σ is exactly λ_i :

$$q_i^T \Sigma q_i = q_i^T (\lambda_i q_i) = \lambda_i.$$

Different eigenportfolios are uncorrelated: $q_i^T \Sigma q_j = 0$ for $i \neq j$. The eigenportfolios form an orthogonal basis in which Σ is diagonal, so we can decompose any portfolio's risk into independent components. On equity universes, the first eigenportfolio q_1 has nearly equal positive weights across all names: it is the market factor. Its eigenvalue λ_1 is much larger than the rest. The second and third eigenportfolios typically correspond to sector contrasts (e.g., long tech, short energy). The bulk of the remaining eigenvalues is sample noise. Numerically, these eigenvectors are computed via the singular value decomposition (SVD) of the returns matrix R rather than forming $R^T R$ directly. SVD generalizes the spectral decomposition to arbitrary matrices: any $R \in \mathbb{R}^{T \times n}$ (even non-square) decomposes as $R = U \Sigma_R V^T$ with orthonormal columns in U and V and a diagonal Σ_R , with no requirement that $T = n$. Forming $R^T R$ explicitly would square the condition number; SVD avoids this.

§5.3 Principal Component Analysis (PCA) and Statistical Factor Models

The top k eigenportfolios form a statistical factor model: $r_i \approx \sum_{j=1}^k \beta_{ij} f_j + \varepsilon_i$, where each factor $f_j = q_j$ is selected from the ordered set $\{q_1, \dots, q_n\}$ of all eigenportfolios, taking only the first k (those with the largest eigenvalues), and β_{ij} are the weights on each factor. The intuition: instead of modeling all $n(n+1)/2$ pairwise covariances in Σ directly, you compress the shared variation across assets into k latent directions and treat the remainder as independent noise. This reduces the parameter count to nk weights plus a diagonal residual, tractable even when n is large.

Statistical factor models are data-driven: the factors emerge from the return data itself, with no prior assumption about what they represent. Whereas **fundamental** models (Fama-French 1993, BARRA) pick observable factors in advance (value, size, momentum, sector), which are interpretable by construction but explain less variance per factor.

§5.4 The Spiked Covariance Model

If the true Σ has a few large eigenvalues above the noise floor, they appear as detectable spikes in the sample spectrum. Below the Baik-Ben Arous-Péché (BBP) threshold $\lambda > 1 + \sqrt{q}$, a true signal disappears into the bulk and is unrecoverable. This is the formal statement of “you cannot tell signal from noise in a high-dimensional covariance.”

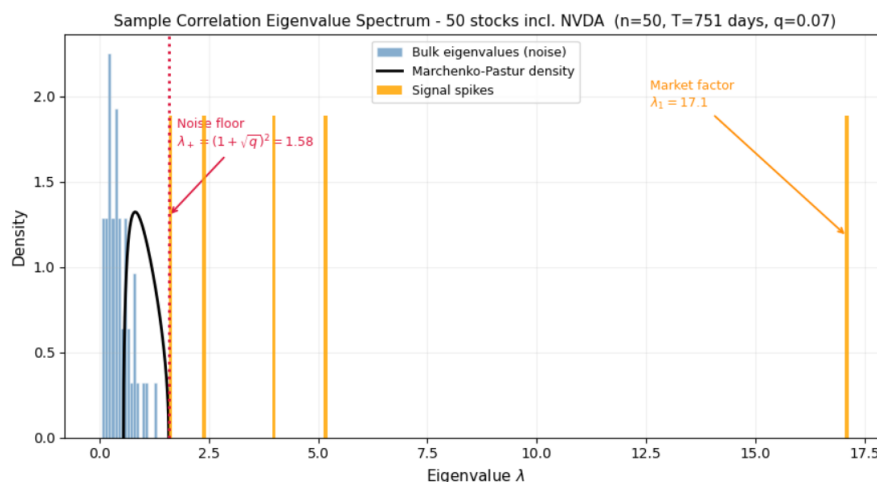


Figure 2: Eigenvalue spectrum of the sample correlation matrix for 50 large-cap stocks including NVDA (2022–2024, $n = 50$, $T = 751$ days, $q = 0.067$). The bulk eigenvalues (blue) follow the Marchenko-Pastur density (black curve); the noise floor $\lambda_+ = (1 + \sqrt{q})^2 = 1.58$ is marked with a dotted red line. Five orange bars above the floor are signal spikes: the market factor dominates at $\lambda_1 = 17.1$, with smaller spikes corresponding to sector contrasts. (See [implementation notebook](#))

§5.5 Ledoit-Wolf Shrinkage

Sample-covariance eigenvalues are spread out compared to the true eigenvalues: large ones are biased upward, small ones biased downward. To see why inverting S is dangerous, recall that $\Sigma = Q\Lambda Q^T$ and $Q^{-1} = Q^T$ (because Q is orthogonal), so

$$\Sigma^{-1} = Q\Lambda^{-1}Q^T.$$

Inversion simply reciprocates each eigenvalue: $\lambda_i \mapsto 1/\lambda_i$. When $n \approx T$, the sample covariance has eigenvalues close to zero, and their reciprocals blow up.

This instability is measured by the **condition number** $\kappa = \lambda_{\max}/\lambda_{\min}$, the ratio of the largest to smallest eigenvalue (equivalently, largest to smallest singular value for a symmetric matrix). A large κ means small perturbations in S produce large changes in S^{-1} , which is exactly Michaud’s error-amplification in matrix form.

Ledoit and Wolf (2004) propose

$$\Sigma_{\text{LW}} = \delta F + (1 - \delta)S,$$

where F is a structured target (commonly constant-correlation) and $\delta \in [0, 1]$ is the shrinkage intensity chosen to minimize expected squared loss against the true Σ . Shrinkage compresses the eigenvalue spectrum toward its mean: the smallest eigenvalue of Σ_{LW} is at least $\Omega(\delta)$, so its reciprocal is $O(1/\delta)$ rather than unbounded. The condition number stays controlled and Σ_{LW}^{-1} does not blow up. Eigenvalue clipping, setting every below-threshold sample eigenvalue to a constant, achieves the same regularization directly in the spectrum.

§5.6 Risk Decomposition via Eigenportfolios

For any portfolio w , write $c = Q^T w$ (coordinates in the eigenportfolio basis). Then

$$w^T \Sigma w = \sum_i c_i^2 \lambda_i.$$

A portfolio's total variance is the sum of its squared eigenportfolio weights (c_i^2) scaled by the eigenvalues. Market-neutral portfolios are those with $c_1 \approx 0$: removing the market factor's contribution means the portfolio's return is uncorrelated with the market. Note that the returns r decomposed here are raw returns, not returns in excess of the risk-free rate r_f . Pure idiosyncratic portfolios are those with $c_i \approx 0$ for all i in the spiked part of the spectrum.

§6 Better Risk-Adjusted Metrics

§6.1 Sortino

Replace standard deviation with downside deviation:

$$\text{Sortino} = (\mu_p - r_f)/\sigma_D, \quad \sigma_D^2 = \mathbb{E}[\min(r - r_f, 0)^2].$$

Sortino penalizes only the part of volatility that hurts. Two strategies with identical Sharpe can have wildly different Sortino: positively skewed returns look much better under Sortino, which matches how investors actually feel about them. A rolling-window Sharpe plot can reveal whether a strategy's edge is stable across regimes or concentrated in a few lucky years.

§6.2 Probabilistic Sharpe Ratio

Definition 6.1. Given a backtest with observed Sharpe $\widehat{\text{SR}}$ over n observations, sample skewness γ_3 , and excess kurtosis γ_4 , the **Probabilistic Sharpe Ratio** (Bailey and Lopez de Prado 2012) for a benchmark SR^* is

$$\text{PSR}(\text{SR}^*) = \Phi \left(\frac{(\widehat{\text{SR}} - \text{SR}^*)\sqrt{n-1}}{\sqrt{1 - \gamma_3 \widehat{\text{SR}} + \frac{\gamma_4 - 1}{4} \widehat{\text{SR}}^2}} \right),$$

where $\Phi(z) = \mathbb{P}(Z \leq z)$ with $Z \sim \mathcal{N}(0, 1)$ is the standard normal CDF, and SR^* is the benchmark Sharpe being tested against.

PSR is a statistical test for whether the true Sharpe ratio exceeds a benchmark. The numerator $(\widehat{\text{SR}} - \text{SR}^*)\sqrt{n-1}$ has exactly the structure of a t -statistic: it counts how many standard errors the observed Sharpe sits above SR^* . The denominator corrects for non-normality of returns: skewness γ_3 and excess kurtosis γ_4 inflate the sampling variance of $\widehat{\text{SR}}$, and the denominator deflates the statistic to account for that. Under normal returns ($\gamma_3 = 0, \gamma_4 = 0$), the denominator equals 1 and the formula reduces to a standard t -test. For $n \geq 30$, the t -distribution with $n-1$ degrees of freedom is well-approximated by $\mathcal{N}(0, 1)$, which is why we apply Φ rather than the t CDF. A high Sharpe based on 50 observations of left-skewed, fat-tailed returns produces a much lower PSR than the same Sharpe based on 5000 observations of well-behaved returns. ([Source: QuantConnect](#))

§6.3 Mean-CVaR

Definition 6.2. For a confidence level $\alpha \in (0, 1)$, the **Value-at-Risk** $\text{VaR}_\alpha(r)$ is the loss threshold exceeded only in the worst α fraction of outcomes:

$$\text{VaR}_\alpha(r) = \inf\{\ell \in \mathbb{R} : \mathbb{P}(-r \geq \ell) \leq \alpha\}.$$

In plain terms: with probability $1 - \alpha$ your loss will not exceed VaR_α . The **Conditional Value-at-Risk** (CVaR) is the average loss given that you are already past that threshold:

$$\text{CVaR}_\alpha(r) = \mathbb{E}[-r \mid -r \geq \text{VaR}_\alpha(r)].$$

In plain terms: CVaR answers “how bad is it on average, given that we are already having a bad day?”

Replace variance with CVaR as the portfolio risk objective. Rockafellar and Uryasev (2000) showed that minimizing CVaR is a linear program in the scenario-based formulation, which makes it tractable at scale. CVaR-aware portfolios behave very differently from variance-aware portfolios in fat-tailed regimes: CVaR sees the tail, variance does not.

§6.4 Which Metric to Use

Use these metrics in combination to form your assessment. None of them are guarantees: they are statistical evidence from your backtest, describing how the strategy behaved in the past under the conditions it faced. A strategy that looks strong across multiple metrics provides more evidence than one that optimizes a single number.

§7 Extra: Why Options Change the Math

For a stock portfolio the P&L is linear in returns: $\text{PnL} = w^T r$. For an option portfolio, the instantaneous change in P&L over a small time step dt is nonlinear

in the underlying:

$$d(\text{PnL}_i) \approx \Delta_i \cdot dS + \frac{1}{2}\Gamma_i \cdot (dS)^2 + \nu_i \cdot d\sigma + \Theta_i \cdot dt,$$

where $dS = S_{t+dt} - S_t$ is the price change over the time step dt , $d\sigma$ is the change in implied volatility, and dt is the length of the time step. The option Greeks are $\Delta = \partial V / \partial S$ (sensitivity of option value V to price), $\Gamma = \partial^2 V / \partial S^2$ (second-order price sensitivity), $\nu = \partial V / \partial \sigma$ (vega, sensitivity to implied volatility), and $\Theta = \partial V / \partial t$ (time decay, negative for long options). The $(dS)^2$ term arises because $V(S)$ is nonlinear: Taylor-expanding $V(S_{t+dt})$ around S_t gives a first-order term $\Delta \cdot dS$ and a second-order correction $\frac{1}{2}\Gamma \cdot (dS)^2$. For a stock, $V = S$ so the second derivative is zero; for an option it is not. Two things change as a result.

First, mean-variance is no longer the right framework. The portfolio P&L distribution is not characterized by mean and variance because the underlying return enters nonlinearly. Even if the stock distribution were Gaussian, the option-portfolio distribution is not.

Second, options give us a degree of freedom equities do not have: **volatility itself**. We can take positions in σ via straddles, strangles, calendars, and variance swaps. The cash market does not offer this. From a portfolio-construction standpoint, options give access to payoff shapes (concave, convex, kinked, bounded) that no linear combination of stocks can replicate, and to volatility as a tradable factor.

§8 Extra: Options Portfolio Optimization

The optimization problem changes shape. We no longer solve $\max w^T \mu - \lambda w^T \Sigma w$ (where $\lambda > 0$ is the risk-aversion parameter from Section 3.2). We solve, schematically,

$$\max_w \frac{\mathbb{E}[\text{PnL}(w; S_T, \sigma_T)]}{\text{Risk}(w; S_T, \sigma_T)}$$

subject to constraints on Greeks, margin, liquidity, and concentration, where S_T is the underlying price and σ_T the implied volatility at the target horizon T . The pipeline is:

1. **Universe construction.** A few hundred candidate positions: single calls and puts, vertical spreads, straddles and strangles, iron condors, calendars, flies, dispersion baskets, earnings-vol trades.
2. **Scenario generation.** Joint distribution over (S_T, σ_T) at horizon T . The scenarios may come from a parametric model (stochastic volatility, jumps), from a bootstrap of historical moves, or from a regime mixture.
3. **Repricing.** Each option in the universe is repriced under every scenario, producing a P&L matrix $L \in \mathbb{R}^{N_{\text{scen}} \times N_{\text{opt}}}$.
4. **Objective.** Compute $\mathbb{E}[\text{PnL}]$, CVaR, or any tail metric directly from L and the weight vector w .

5. **Constraints.** Greek-neutrality at the portfolio level (target $\Delta = 0$ or bounded Γ , ν within limits), margin caps, position-size limits, turnover limits, liquidity (bid-ask, open interest).
6. **Solve.** The problem is generally non-convex (because of mixed positions, integer lots, or scenario-dependent constraints). Common formulations: stochastic programming with sample-average approximation, convex relaxations, mixed-integer programs when whole-contract sizes matter, or robust formulations when scenario probabilities are themselves uncertain.
7. **Hedge dynamically.** Once on, the portfolio's Greeks drift as the underlying and vol surface move. Continuous (or periodic) delta and vega hedging is what keeps the optimized exposure on target.

§9 What We Haven't Talked About

A non-exhaustive list of topics that belong in a longer course:

- **Risk parity** and inverse-volatility allocation (Qian, Bridgewater All Weather).
- **Regime-switching models:** hidden Markov models, Markov-switching GARCH.
- **Machine-learning return forecasts:** tree ensembles, deep learning on alternative data, the over-fitting problem at scale.
- **Transaction-cost models and execution:** Almgren-Chriss optimal execution, market-impact models.
- **Robust optimization:** uncertainty sets on μ and Σ , worst-case Sharpe.
- **Performance attribution** at production scale (Brinson, factor-based, residual).

Each of these is a course of its own. What this lecture aimed to give you is a coherent through-line: why we build portfolios, why the obvious answer (Markowitz) is incomplete, what shape the corrections take (factor models, shrinkage, better metrics), and how all of it bends when payoffs become nonlinear.

References

- [1] H. Markowitz, “Portfolio Selection,” *Journal of Finance*, 1952.
- [2] W. F. Sharpe, “Capital Asset Prices: A Theory of Market Equilibrium Under Conditions of Risk,” *Journal of Finance*, 1964.
- [3] R. O. Michaud, “The Markowitz Optimization Enigma: Is ‘Optimized’ Optimal?,” *Financial Analysts Journal*, 1989.
- [4] F. Black and R. Litterman, “Global Portfolio Optimization,” *Financial Analysts Journal*, 1992.
- [5] O. Ledoit and M. Wolf, “Honey, I Shrunk the Sample Covariance Matrix,” *Journal of Portfolio Management*, 2004.
- [6] F. A. Sortino and L. N. Price, “Performance Measurement in a Downside Risk Framework,” *Journal of Investing*, 1994.
- [7] D. H. Bailey and M. Lopez de Prado, “The Sharpe Ratio Efficient Frontier,” *Journal of Risk*, 2012.
- [8] R. T. Rockafellar and S. Uryasev, “Optimization of Conditional Value-at-Risk,” *Journal of Risk*, 2000.
- [9] G. A. Paleologo, *The Elements of Quantitative Investing*. Hoboken, NJ: John Wiley & Sons (Wiley Finance), 2025.
- [10] J. Simonson & D. Melchin, “Probabilistic Sharpe Ratio,” QuantConnect Research, 2024. <https://www.quantconnect.com/research/17112/probabilistic-sharpe-ratio/>
- [11] R. Paolucci, *Quant Guild* [YouTube channel]. <https://www.youtube.com/@QuantGuild>
- [12] Sigma Squared, “The Math of the ‘Perfect’ Portfolio” [video]. <https://www.youtube.com/watch?v=4hXKYzReGyE>